Last 20 Scoring Numbers

Here are the last 20 numbers again for reference:

- 90
- 93
- 101
- 91
- 90
- 94
- 95
- 96
- 94
- 89
- 97
- 93
- 90
- 85
- 94
- 85
- 91
- 116
- 93
- 93
- 98
- 89
- 99
- 85

Sorting the Numbers

Now, let's sort these numbers in ascending order to identify the lowest values:

1. 85 (occurs three times)
2. 89 (occurs twice)
3. 90 (occurs three times)
4. 91 (occurs twice)
5. 93 (occurs four times)
6. 94 (occurs three times)
7. 95
8. 96
9. 97
10. 98
11. 99
12. 101
13. 116

Lowest 8 Scores

The lowest 8 numbers from the last 20 are:

1. 85
2. 85

3. 85
4. 89
5. 89
6. 90
7. 90
8. 90

Using the PGA handicap system what is the handicap based on those 8 numbers, course rating of 71.5 and slope rating of 113

Given Data

- **Scores**: 85, 85, 85, 89, 89, 90, 90, 90
- **Course Rating**: 71.5
- **Slope Rating**: 113

Step-by-Step Calculation

1. **Calculate Handicap Differentials for Each Score**:
   The formula for the Handicap Differential is:
3. $$\text{Handicap Differential} = \frac{(\text{Adjusted Gross Score} - \text{Course Rating}) \times 113}{\text{Slope Rating}}$$
5. Since we are using the gross scores directly (assuming no adjustments), we can plug in the values:
   - For **85**:
     $$\text{Handicap Differential} = \frac{(85 - 71.5) \times 113}{113} = (85 - 71.5) = 13.5$$
   - For **89**:
     $$\text{Handicap Differential} = (89 - 71.5) = 17.5$$
   - For **90**:
     $$\text{Handicap Differential} = (90 - 71.5) = 18.5$$

7. Now calculating for each score:
   - **85**: 13.5
   - **85**: 13.5
   - **85**: 13.5
   - **89**: 17.5
   - **89**: 17.5
   - **90**: 18.5
   - **90**: 18.5
   - **90**: 18.5
2. **List of Handicap Differentials**:
   The differentials are:
   - $13.5, 13.5, 13.5, 17.5, 17.5, 18.5, 18.5, 18.5$
3. **Select Lowest Handicap Differentials**:
   Since we have only eight scores, we will use all of them.
4. **Average the Lowest Handicap Differentials**:
   Calculate the average of these differentials:

$$\text{Average} = \frac{(13.5 + 13.5 + 13.5 + 17.5 + 17.5 + 18.5 + 18.5 + 18.5)}{8}$$

Summing these values:

$$= (13.5 + 13.5 + 13.5 + 17.5 + 17.5 + 18.5 + 18.5 + 18.5) = 130$$

Now divide by the number of scores:

$$= \frac{130}{8} = 16.25$$

5. **Multiply by 0.96 (96%)**:

$$= 16.25 \times 0.96 = 15.6$$

6. **Truncate to Tenths Place**:

The final step is to truncate to the tenths place:

Thus, the **Handicap Index** is 15.

7. **Calculate Course Handicap**:
   Using the Course Handicap formula:

$$\text{Course Handicap} = \text{Handicap Index} \times \left(\frac{\text{Slope Rating}}{113}\right)$$

Plugging in the values:

$$= 15 \times \left(\frac{113}{113}\right) = 15$$

Final Result

The calculated Course Handicap based on the provided scores is **15**.